Integer Lessons and Adaptive Question System

Grade 7 Mathematics — Curriculum Master

Three lessons. Six skills. 64 questions outside the protected pre/post set.

Lesson 1 — Understanding and representing integers

Skills: S1 + S5

Objectives:
1. Identify integers, opposites, and absolute value.
2. Represent, compare, and order integers on a number line.

What is an integer?

An integer is a whole number. It can be positive, negative, or zero. It has no fractional part and no decimal.

Examples: -8, -3, 0, 5, 12.
Not integers: 3/4, 0.6, 12%.

We use negative numbers every day. A temperature 3 degrees below zero is -3. A debt of 500 pesos is -500. A diver 6 meters under the sea is at -6.

The number line

A number line shows integers in order. Zero is in the middle. Positive integers go right. Negative integers go left.

Values increase as you move right. So $-7 < -2 < 0 < 4$.

Compare by position, not by digit size. -7 is smaller than -2 because it is farther left.

[insert figure: lesson1/integer-number-line]

Opposites and absolute value

Opposites are two integers the same distance from zero on different sides. -5 and 5 are opposites.

Absolute value is the distance from zero. Distance is always positive or zero. We write it with two bars.

- $|5| = 5$

- |-5| = 5
- |0| = 0

[insert figure: lesson1/opposites-example]

[insert video: VID-INTEGER-NUMBER-LINE | 1:00-1:50] (optional)

Worked example

Problem: Is -11 an integer? What is its opposite? What is |-11|?

Step 1: -11 has no fraction and no decimal. So it is an integer.
Step 2: The opposite of -11 is 11. Same distance, other side of zero.
Step 3: |-11| = 11 because -11 is 11 units from zero.
Answer: -11 is an integer. Opposite: 11. Absolute value: 11.

Common mistake

-7 > -3 is wrong. The digit 7 is bigger, but -7 is farther left. Correct: -7 < -3.

Ready to practice

Continue to the regular exercise for 6 questions covering both skills.

Lesson 1 — Regular exercise (6 questions)

L1-R-S1-01 | S1 | Regular | text | Bloom: Understand (LOTS)

Which value is an integer?
A. -8
B. 3/4
C. 0.6
D. 12%

Answer: A. -8 has no fractional part.

L1-R-S1-02 | S1 | Regular | text | Bloom: Apply (LOTS)

What is the opposite of -12?
A. -12
B. 0
C. 12
D. 1/12

Answer: C. Same distance from zero, different sign.

L1-R-S1-03 | S1 | Regular | figure | Bloom: Analyze (HOTS)

What is |-9|?

[insert figure: L1-R-S1-03]

A. -9
B. 9
C. 0
D. 1/9

Answer: B. The distance from -9 to zero is 9.

L1-R-S5-01 | S5 | Regular | text | Bloom: Understand (LOTS)

Which list is ordered least to greatest?
A. -6, -2, 0, 4
B. 4, 0, -2, -6
C. -2, -6, 0, 4
D. -6, 0, -2, 4

Answer: A. Values increase left to right.

L1-R-S5-02 | S5 | Regular | text | Bloom: Apply (LOTS)

Which comparison is true?
A. -7 > -3
B. -7 = -3
C. -7 < -3
D. -3 < -7

Answer: C. -7 is to the left of -3.

L1-R-S5-03 | S5 | Regular | figure | Bloom: Analyze (HOTS)

Which plotted value is greatest?

[insert figure: L1-R-S5-03]

A. -6
B. -1
C. 3
D. 6

Answer: D. 6 is farthest right.

Lesson 1 — Additional exercise (6 questions)

L1-A-S1-01 | S1 | Additional | text | Bloom: Understand (LOTS)

Which value belongs to the integers?
A. -15
B. 2.5
C. 1/3
D. 7.1%

Answer: A. -15 is a whole number.

L1-A-S1-02 | S1 | Additional | figure | Bloom: Apply (LOTS)

The point is -4. Which value is its opposite?

[insert figure: L1-A-S1-02]

A. -4
B. 0
C. 4
D. 8

Answer: C. Reflect -4 across zero.

L1-A-S1-03 | S1 | Additional | text | Bloom: Analyze (HOTS)

What is |-14|?
A. -14
B. 0
C. 14
D. 28

Answer: C. Distance from zero is 14 units.

L1-A-S5-01 | S5 | Additional | text | Bloom: Understand (LOTS)

Order -1, 5, -5, and 0 from greatest to least.
A. 5, 0, -1, -5
B. -5, -1, 0, 5
C. 5, -1, 0, -5
D. 0, 5, -1, -5

Answer: A. Start with the farthest right.

L1-A-S5-02 | S5 | Additional | text | Bloom: Apply (LOTS)

Which comparison is correct?
A. -2 < -8
B. -8 > 0
C. -2 > -8
D. 0 < -8

Answer: C. -2 lies right of -8.

L1-A-S5-03 | S5 | Additional | figure | Bloom: Analyze (HOTS)

Which is the middle plotted value?

[insert figure: L1-A-S5-03]

A. -4
B. 0
C. 7
D. -7

Answer: B. 0 lies between -4 and 7.

Lesson 1 — Foundational exercise (S1: 3 questions)

L1-F-S1-01 | S1 | Foundational | text | Bloom: Remember (LOTS)

Zero belongs to which set?
A. Integers
B. Only fractions
C. Only decimals
D. Only percentages

Hint: Integers include negatives, zero, and positive whole numbers.
Answer: A. Zero is an integer.

L1-F-S1-02 | S1 | Foundational | text | Bloom: Understand (LOTS)

What is the opposite of 5?
A. 5
B. -5

C. 0
D. 10

Hint: Keep the distance, change the sign.
Answer: B. Mirror 5 across zero.

L1-F-S1-03 | S1 | Foundational | figure | Bloom: Analyze (HOTS)

What is |-3|?

[insert figure: L1-F-S1-03]

A. -3
B. 0
C. 3
D. 6

Hint: Count steps from -3 to zero.
Answer: C. Three unit steps to zero.

Lesson 1 — Foundational exercise (S5: 3 questions)

L1-F-S5-01 | S5 | Foundational | text | Bloom: Remember (LOTS)

Which is greater, -2 or -5?
A. -5
B. -2
C. They are equal
D. 0

Hint: The rightmost number is greater.
Answer: B. -2 is farther right.

L1-F-S5-02 | S5 | Foundational | figure | Bloom: Understand (LOTS)

Order -3, 0, and 2 from least to greatest.

[insert figure: L1-F-S5-02]

A. 2, 0, -3
B. -3, 2, 0
C. -3, 0, 2
D. 0, -3, 2

Hint: Begin with the negative value.
Answer: C. Read left to right.

L1-F-S5-03 | S5 | Foundational | text | Bloom: Analyze (HOTS)

Which symbol completes -4 __ 1?
A. <
B. >
C. =
D. +

Hint: A negative number is left of a positive number.
Answer: A. -4 is smaller than 1.

Lesson 2 — Operations with integers

Skills: S2 + S3

Objectives:
1. Add and subtract integers.
2. Multiply and divide integers.

Zero pairs

When you add integers with different signs, a positive chip and a negative chip cancel. Each pair equals zero.

Example: -5 + 3.
Start with 5 negative chips. Add 3 positive chips. Three pairs cancel. Two negatives remain.
-5 + 3 = -2.

[insert figure: lesson2/chip-model-zero-pairs]

Adding same signs

Same signs: add the distances and keep the sign.

- (-4) + (-3) = -7
- 4 + 3 = 7

Subtracting

Subtracting an integer is the same as adding its opposite.

a - b = a + (-b)

Example: 4 - (-3) = 4 + 3 = 7.

Multiplying and dividing

Work with distances first. Then decide the sign.

- Same signs -> positive
- Different signs -> negative

Examples:
- (-6) x 4 = -24
- (-6) x (-4) = 24
- (-24) / (-6) = 4
- 24 / (-6) = -4

[insert figure: lesson2/sign-rules]

[insert video: VID-INTEGER-ADD-SUBTRACT | 1:20-2:05] (optional)

[insert video: VID-INTEGER-MULTIPLY-DIVIDE | 0:45-1:30] (optional)

Worked example

Problem: Compute (-6) x 4 and (-24) / (-6).

Step 1: Signs of (-6) x 4 are different. Result is negative. 6 x 4 = 24.
Answer 1: -24.

Step 2: Signs of (-24) / (-6) are the same. Result is positive. 24 / 6 = 4.
Answer 2: 4.

Common mistake

(-6) x 4 = 24 is wrong. The signs are different, so the answer is -24.

Ready to practice

Continue to the regular exercise for 6 questions.

Lesson 2 — Regular exercise (6 questions)

L2-R-S2-01 | S2 | Regular | text | Bloom: Understand (LOTS)

Compute -18 + (-7).
A. -25
B. -11
C. 11
D. 25

Answer: A. Add distances, keep the negative sign.

L2-R-S2-02 | S2 | Regular | figure | Bloom: Apply (LOTS)

Use the arrow to compute -9 + 14.

[insert figure: L2-R-S2-02]

A. -23
B. -5
C. 5
D. 23

Answer: C. Fourteen steps right from -9 lands at 5.

L2-R-S2-03 | S2 | Regular | text | Bloom: Analyze (HOTS)

Compute -31 - (-12).
A. -43
B. -19
C. 19
D. 43

Answer: B. Subtracting -12 means adding 12.

L2-R-S3-01 | S3 | Regular | text | Bloom: Understand (LOTS)

Compute (-7) x 6.
A. -42
B. 42
C. -13
D. 13

Answer: A. Different signs give a negative result.

L2-R-S3-02 | S3 | Regular | text | Bloom: Apply (LOTS)

Compute (-48) / (-8).
A. -6
B. 6

C. -40
D. 40

Answer: B. Same signs give a positive result.

L2-R-S3-03 | S3 | Regular | figure | Bloom: Analyze (HOTS)

What do four equal groups of -3 total?

[insert figure: L2-R-S3-04]

A. -12
B. 12
C. -7
D. 7

Answer: A. 4 x (-3) = -12.

Lesson 2 — Additional exercise (6 questions)

L2-A-S2-01 | S2 | Additional | text | Bloom: Understand (LOTS)

Compute -24 + 9.
A. -33
B. -15
C. 15
D. 33

Answer: B. The larger distance is negative.

L2-A-S2-02 | S2 | Additional | figure | Bloom: Apply (LOTS)

Compute 17 - (-6).

[insert figure: L2-A-S2-02]

A. 11
B. -11
C. 23
D. -23

Answer: C. Subtracting -6 means adding 6.

L2-A-S2-03 | S2 | Additional | text | Bloom: Analyze (HOTS)

Compute -42 - (-19).
A. -61
B. -23
C. 23
D. 61

Answer: B. -42 + 19 = -23.

L2-A-S3-01 | S3 | Additional | text | Bloom: Understand (LOTS)

Compute (-9) x (-4).
A. -36
B. 36
C. -13
D. 13

Answer: B. Same signs give a positive product.

L2-A-S3-02 | S3 | Additional | text | Bloom: Apply (LOTS)

Compute (-63) / 7.
A. -9
B. 9
C. -56
D. 56

Answer: A. Different signs give a negative quotient.

L2-A-S3-03 | S3 | Additional | figure | Bloom: Analyze (HOTS)

Compute 96 / (-12).

[insert figure: L2-A-S3-01]

A. -8
B. 8
C. -84
D. 84

Answer: A. 96 / 12 = 8, signs differ.

Lesson 2 — Foundational exercise (S2: 3 questions)

L2-F-S2-01 | S2 | Foundational | text | Bloom: Remember (LOTS)

Compute -4 + (-3).
A. -7
B. -1
C. 1
D. 7

Hint: Two negatives combine into a larger negative.
Answer: A. Move 3 left from -4.

L2-F-S2-02 | S2 | Foundational | figure | Bloom: Understand (LOTS)

Compute -6 + 8.

[insert figure: L2-F-S2-02]

A. -14
B. -2
C. 2
D. 14

Hint: Pair each negative with one positive.
Answer: C. Six pairs cancel, two positives remain.

L2-F-S2-03 | S2 | Foundational | text | Bloom: Analyze (HOTS)

Compute 3 - (-2).
A. 1
B. -1
C. 5
D. -5

Hint: Rewrite as 3 + 2.
Answer: C. Subtract negative = add positive.

Lesson 2 — Foundational exercise (S3: 3 questions)

L2-F-S3-01 | S3 | Foundational | text | Bloom: Remember (LOTS)

Compute (-3) x 4.
A. -12
B. 12
C. -7

D. 7

Hint: Find 3 x 4 first, then the sign.
Answer: A. One negative factor gives a negative product.

L2-F-S3-02 | S3 | Foundational | text | Bloom: Understand (LOTS)

Compute (-12) / 3.
A. -4
B. 4
C. -9
D. 9

Hint: 12 / 3 = 4, signs differ.
Answer: A. Different signs make a negative quotient.

L2-F-S3-03 | S3 | Foundational | figure | Bloom: Analyze (HOTS)

Compute (-2) x (-5).

[insert figure: L2-F-S3-03]

A. -10
B. 10
C. -7
D. 7

Hint: Both factors are negative.
Answer: B. Same signs -> positive product.

Lesson 3 — Application of integers

Skills: S4 + S6

Objectives:
1. Evaluate expressions using order of operations.
2. Model signed-change situations.

A story about temperature

The temperature at 7:00 AM was -7 degrees C. By noon it rose by 8 degrees.

The change is +8. Add: -7 + 8 = 1.
The new temperature is 1 degree C.

Order of operations

When an expression has more than one operation:

1. Grouping first: ( ), [ ], { }.
2. Multiply and divide, left to right.
3. Add and subtract, left to right.

[insert figure: lesson3/operation-order-steps]

Words that mean signs

Rise, gain, deposit, up -> positive.
Fall, loss, withdrawal, down -> negative.

Interpreting the result

Always include the unit. Say if the result is positive or negative. Explain what it means.

[insert figure: lesson3/temperature-change]

[insert video: VID-INTEGER-ORDER-OPERATIONS | 0:40-1:25] (optional)

Worked example

Problem: Evaluate -3 x (-4) + (-18) / 6 - (-5).

Step 1: Do multiplication and division first, left to right.
-3 x (-4) = 12
(-18) / 6 = -3
Step 2: Rewrite: 12 + (-3) - (-5).
Step 3: Subtract negative = add positive: 12 + (-3) + 5.
Step 4: Add left to right: 12 - 3 = 9, then 9 + 5 = 14.
Answer: 14.

Common mistake

Adding before dividing is wrong. In -3 x (-4) + (-18) / 6, division goes before addition. Correct order gives 14.

Ready to practice

Continue to the regular exercise for 6 questions.

Lesson 3 — Regular exercise (6 questions)

L3-R-S4-01 | S4 | Regular | figure | Bloom: Understand (LOTS)

Evaluate 5 + (-3) x 4.

[insert figure: L3-R-S4-01]

A. 8
B. -7
C. -32
D. -20

Answer: B. Multiply first: -12, then 5 + (-12) = -7.

L3-R-S4-02 | S4 | Regular | text | Bloom: Apply (LOTS)

Evaluate 18 / (-3) + 7.
A. -13
B. -1
C. 1
D. 13

Answer: C. Divide first to get -6, then -6 + 7 = 1.

L3-R-S4-03 | S4 | Regular | text | Bloom: Analyze (HOTS)

Evaluate (-2) x [6 - 9].
A. -6
B. 6
C. -18
D. 18

Answer: B. The bracket is -3; (-2) x (-3) = 6.

L3-R-S6-01 | S6 | Regular | figure | Bloom: Understand (LOTS)

It is -4 degrees C and warms by 9 degrees. What is the new temperature?

[insert figure: L3-R-S6-01]

A. -13
B. -5
C. 5
D. 13

Answer: C. -4 + 9 = 5.

L3-R-S6-02 | S6 | Regular | text | Bloom: Apply (LOTS)

A diver is at -6 m and descends 4 m. What is the new position?
A. -10 m
B. -2 m
C. 2 m
D. 10 m

Answer: A. -6 + (-4) = -10.

L3-R-S6-03 | S6 | Regular | text | Bloom: Analyze (HOTS)

An elevator starts at floor 3 and goes down 7 floors. Where does it stop?
A. -10
B. -4
C. 4
D. 10

Answer: B. 3 - 7 = -4.

Lesson 3 — Additional exercise (6 questions)

L3-A-S4-01 | S4 | Additional | figure | Bloom: Understand (LOTS)

Evaluate (-4) x (-5) - 27 / 9.

[insert figure: L3-A-S4-01]

A. 17
B. 23
C. -17
D. -23

Answer: A. 20 - 3 = 17.

L3-A-S4-02 | S4 | Additional | text | Bloom: Apply (LOTS)

Evaluate 8 - [3 + (-7)].
A. -2
B. 4
C. 12
D. 18

Answer: C. The bracket is -4; 8 - (-4) = 12.

L3-A-S4-03 | S4 | Additional | text | Bloom: Analyze (HOTS)

Evaluate (-24) / 6 x (-2).
A. -8
B. 8
C. -2
D. 2

Answer: B. Left to right: -4 x (-2) = 8.

L3-A-S6-01 | S6 | Additional | figure | Bloom: Understand (LOTS)

A submarine is at -18 m and rises 7 m. Where is it now?

[insert figure: L3-A-S6-01]

A. -25 m
B. -11 m
C. 11 m
D. 25 m

Answer: B. -18 + 7 = -11.

L3-A-S6-02 | S6 | Additional | text | Bloom: Apply (LOTS)

The temperature is 6 degrees C and falls 14 degrees. What is the new temperature?
A. -20
B. -8
C. 8
D. 20

Answer: B. 6 - 14 = -8.

L3-A-S6-03 | S6 | Additional | text | Bloom: Analyze (HOTS)

A hiker is 12 m above sea level, descends 19 m, then rises 3 m. Where is the hiker?
A. -10 m
B. -4 m
C. 4 m
D. 10 m

Answer: B. 12 - 19 + 3 = -4.

Lesson 3 — Foundational exercise (S4: 3 questions)

L3-F-S4-01 | S4 | Foundational | figure | Bloom: Remember (LOTS)

Evaluate 2 + 3 x 4.

[insert figure: L3-F-S4-01]

A. 14
B. 20
C. 24
D. 9

Hint: Multiply first.
Answer: A. 2 + 12 = 14.

L3-F-S4-02 | S4 | Foundational | text | Bloom: Understand (LOTS)

Evaluate (5 - 8) x (-2).
A. -6
B. 6
C. -26
D. 26

Hint: Do the bracket first.
Answer: B. Bracket is -3; (-3) x (-2) = 6.

L3-F-S4-03 | S4 | Foundational | text | Bloom: Analyze (HOTS)

Evaluate 12 / (-3) + 1.
A. -5
B. -3
C. 3
D. 5

Hint: Divide first.
Answer: B. -4 + 1 = -3.

Lesson 3 — Foundational exercise (S6: 3 questions)

L3-F-S6-01 | S6 | Foundational | figure | Bloom: Remember (LOTS)

It is -2 degrees C, then warms 5 degrees. What is the temperature?

[insert figure: L3-F-S6-01]

A. -7
B. -3
C. 3
D. 7

Hint: Warming adds a positive change.
Answer: C. -2 + 5 = 3.

L3-F-S6-02 | S6 | Foundational | text | Bloom: Understand (LOTS)

A diver is at -3 m and goes down 2 m. Where is the diver?
A. -5 m
B. -1 m
C. 1 m
D. 5 m

Hint: Down means subtract 2.
Answer: A. -3 - 2 = -5.

L3-F-S6-03 | S6 | Foundational | text | Bloom: Analyze (HOTS)

What expression models a rise of 4 from -6?
A. -6 + 4
B. -6 - 4
C. 6 + 4
D. 6 - 4

Hint: Start value plus change.
Answer: A. A rise adds a positive change.

Whole-course activity (10 questions)

ACT-S1-01 | S1 | Activity | figure | Bloom: Understand (LOTS)

The point is -6. Which integer is its opposite?

[insert figure: ACT-S1-01]

A. -6

B. 0
C. 6
D. 12

Answer: C. Reflect -6 across zero.

ACT-S1-02 | S1 | Activity | text | Bloom: Understand (LOTS)

Which value is an integer?
A. -14
B. 1/2
C. 0.4
D. 40%

Answer: A. -14 has no fractional part.

ACT-S2-01 | S2 | Activity | figure | Bloom: Understand (LOTS)

The model has four negative and seven positive chips. What is the sum?

[insert figure: ACT-S2-01]

A. -11
B. -3
C. 3
D. 11

Answer: C. Four zero pairs cancel; three positives remain.

ACT-S2-02 | S2 | Activity | text | Bloom: Apply (LOTS)

Compute -18 + (-7).
A. -25
B. -11
C. 11
D. 25

Answer: A. Add distances, keep the negative sign.

ACT-S3-01 | S3 | Activity | text | Bloom: Understand (LOTS)

Compute (-7) x 6.
A. -42
B. 42
C. -13
D. 13

Answer: A. Different signs give a negative product.

ACT-S3-02 | S3 | Activity | text | Bloom: Analyze (HOTS)

Compute (-48) / (-8).
A. -6
B. 6
C. -40
D. 40

Answer: B. Same signs give a positive quotient.

ACT-S4-01 | S4 | Activity | text | Bloom: Understand (LOTS)

Evaluate 5 + (-3) x 4.
A. 8
B. -7
C. -32
D. -20

Answer: B. Multiply first: -12, then 5 + (-12) = -7.

ACT-S4-02 | S4 | Activity | text | Bloom: Analyze (HOTS)

Evaluate (-2) x [6 - 9].
A. -6
B. 6
C. -18
D. 18

Answer: B. Bracket is -3; (-2) x (-3) = 6.

ACT-S5-01 | S5 | Activity | video | Bloom: Understand (LOTS)

After the number-line segment, which is greatest: -5, 0, 2, or 7?

[insert video: VID-INTEGER-NUMBER-LINE | 1:00-1:50]

A. -5
B. 0
C. 2
D. 7

Answer: D. 7 is farthest right.

ACT-S6-01 | S6 | Activity | text | Bloom: Apply (LOTS)

It is -4 degrees C and warms by 9 degrees. What is the new temperature?
A. -13
B. -5
C. 5
D. 13

Answer: C. -4 + 9 = 5.

Media manifest

Four approved videos (all optional, watch at home):

1. VID-INTEGER-NUMBER-LINE - "Negative numbers and number line examples" - 1:00-1:50
2. VID-INTEGER-ADD-SUBTRACT - "Adding and subtracting negative numbers" - 1:20-2:05
3. VID-INTEGER-MULTIPLY-DIVIDE - "Multiplying and dividing negative numbers" - 0:45-1:30
4. VID-INTEGER-ORDER-OPERATIONS - "Order of operations example" - 0:40-1:25